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Nil Mamano, PhD

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Experience

Beyond Cracking the Coding Interview

Author & Lead Developer amazon.com/dp/195570600X — 2025

Co-wrote the official sequel to Cracking the Coding Interview with Gayle L. McDowell et al. #1 Best Seller in Data Structures & Algorithms on Amazon.

Built dsatoolkit.com, a companion DS&A toolkit. Built an automated pipeline to parse, test, and generate articles for all 1,440 code solutions across 5 languages. (Next.js, TypeScript, Python)

Built LLM orchestration pipelines for large-scale codebase changes, e.g., translating 288 book solutions from Python to Go with few-shot prompting and test-driven retry loops, using local models and APIs.

Co-designed an AI interviewer for the book problems (bctci.co/ai).

Google

Senior Software Engineer — Feb 2021 – Aug 2024

Worked on Google's software-defined WAN, which carries traffic between all Google data centers. (C++, Go, Python, SQL)

Part of a team that migrated the bandwidth ordering system from a quarterly manual process to an on-demand service with daily resolution, giving internal teams more flexibility and improving WAN utilization.

Built and owned the bandwidth prioritization component. Designed time-series-based algorithms for distributing the available bandwidth among teams when demand exceeds capacity, balancing prioritization, fairness, and continuity constraints.

Part of a cross-functional effort to rapidly provision network capacity for Gemini.

Pathrise

Tech Interview Consultant — Apr 2020 – Jan 2021

Revamped the DS&A curriculum for coding interviews. Taught algorithms to 100+ students.

Education

PhD + Masters in Computer Science University of California Irvine, GPA 3.83/4 — Sep 2015 – Dec 2019

scholar.google.bg/citations?user=LIuligEAAAAJ | Co-authored 9 peer-reviewed papers on algorithm design, including as main author in tier A conferences like ICALP and ISAAC. The papers describe new algorithmic improvements in graph theory, computational geometry, and computational biology.

Led a research project from inception to publication: came up with an original problem, engaged 3 colleagues to work on it, and collaborated with them to solve it and write a paper. We invented an algorithm for the knight's tour problem (cited by Knuth).

Led 100+ sessions and guest lectures with 50+ students. Led a 120-student study on the effect of immediate automated feedback on learning outcomes.

B.E. in Computer Science Polytechnic University of Catalonia, GPA 3.8/4 (99th percentile) — Sep 2011 – Jul 2015

github.com/nmamano/SANA | Created SANA, a C++ biological network alignment tool that uses simulated annealing. 100+ citations; actively maintained for 10+ years by 50+ collaborators, with 2,000+ commits.

Projects

Wall Game wallgame.io — 2025

A full-stack online board game supporting AIs as first-class citizens via an AI integration protocol/client. (React, TypeScript, Tailwind, shadcn, PostgreSQL, MongoDB, WebSocket)

Trained multiple AlphaZero-style models for different board sizes and rule sets via self-play on consumer GPUs. (C++, PyTorch, TensorRT, ONNX, WebAssembly)

Technical Blog nilmamano.com/blog — 2025

30+ articles on algorithms, data structures, AI, CS research, and software engineering.